OO-M-671D
August 3, 1977
SUPERSEDING
Fed. Spec. OO-M-671C
September 24, 1959

FEDERAL SPECIFICATION

MOWER, LAWN, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers standard cutting reel hand lawn mowers that are pushed or propelled by manual effort. It is used for cutting grass ordinarily grown on lawns.

1.2 Classification.

1.2.1 Types, classes, and sizes. The mowers covered by this specification shall be the rotating reel, hand propelled and furnished in the following types, classes, and sizes as specified herein and in the invitation for bids, contract, or order (see 6.2).

Type I - Silent type.
Type II - Scissor type.

Class A - 5 bladed reel.
Class B - 6 bladed reel.

Size 1 - 16 ± 1 inch cutting width.
Size 2 - 18 ± 1 inch cutting width.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

FED. STD. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:00Z
Check the source to verify that this is the current version before use.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

E-18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Standard Methods of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committe, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Condition. Lawn mowers shall be complete and new. Handles need not be assembled or affixed to the machine. There shall be no defects or noticeable wear in any of the component parts of mechanisms upon the conclusion of any test conducted to determine compliance with the service, cutting, or other performance requirements of this specification. Mowers shall be free from rust and imperfections which may affect appearance or serviceability of the mower.

3.1.1 Standard product. The lawn mowers furnished under this specification shall be the manufacturer's current standard, commercial product except for those modifications or changes required in the standard product in order to comply with the requirements of this specification. All accessories and components shall be furnished with each lawn mower that are normally furnished with the standard product offered commercially.

3.2 Materials. Materials used in the construction of lawn mowers shall be of the type, quality, and condition necessary and suitable for the specific application. Recycled material shall be used where applicable.

3.3 Fasteners. Where aluminum-magnesium or zinc alloys are involved, assembly of parts shall be by the following: through bolts, bushings with tapped holes, threaded studs inserted to a depth of two stud diameters minimum, or coil type inserts. All threaded fasteners shall be secured to prevent loosening under vibration in a manner which will permit ready disassembly for maintenance or repair. Swaging, peening, or staking of threads shall not be permitted.

3.4 Design and construction. The mowers shall consist essentially of cutting bar, reel and blades, handle, and supporting structural frame. Castings shall be smooth, sound, and free from blow holes, pits, foreign material, or other defects. Frames and supporting structural members shall be capable of withstanding loads arising from operation on rough grounds without breakage, visible damage or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

misalignment. Components and related parts shall be accurately and permanently aligned. The clearance or opening in any gear housing formed by the side frame and wheel shall not exceed 3/32 inch. Maintenance and operating instructions for maximizing operational life of the lawn mower shall be furnished with each unit (see 3.17).

3.4.1 Blades and bed knife. The cutting blades shall consist of a multi-blade rotating reel and a single blade fixed bed knife, both supported by the structural frame to maintain uniform clearance and alignment. The hardness of the blades of the reel shall be compatible with the associated bed knife when measured within 3/16 inches of each sharpened cutting edge. Reel blades shall be either alloy or high carbon steel. The average hardness shall be not less than Rockwell C38 nor more than C54 after the blades are finish-ground. Blades shall extend beyond the spider sufficiently to permit repeated regrindings. The bed knife shall be either alloy or high carbon tool steel. The average hardness shall be not less than Rockwell C48 nor more than C58 after the knife is finish-ground. Testing for hardness shall be in accordance with 4.4.4.

3.4.1.1 Reels. The cutting reel shall be of rugged and durable construction and the blades shall be precision ground concentric with respect to the shaft. Reels for class A mowers shall have five blades and reels for class B mowers shall have six blades. The reel shaft shall be not less than 0.56 inches in diameter. Reel spiders shall be either malleable iron or steel. Each reel blade shall be attached securely to each spider by either riveting or welding.

3.4.1.2 Backing bar and bed knife. The backing bar and bed knife shall have a screw mechanism for setting proper clearance between the reel and bed knife and shall remain securely locked and shall not creep or loosen under any operating conditions of the lawn mower. The screw mechanism shall be easily accessible so that adjusting tools will not slip out of screw slots or wrenches off of jam nuts. The width of the bed knife shall be sufficient to permit repeated regrinding. For type I mowers the cutting edge of the bed knife shall be chamfered tangent to the circle generated by the rotation of the reel blades. For the type II mowers the cutting edge of the bed knife shall be ground with an interior angel of less than 90 degrees. The cutting edge shall be straight and true when held against a straight edge or flat surface. The bed knife shall be stress relieved after grinding to prevent future warping. The bed knife shall be flat when viewed from either end without any dip or roll-off. The bed knife may be one piece or two-piece construction with separate bed knife and backing bar. If two-piece construction, the portion of the surface of the backing bar making contact with the bed knife shall be finished (rough ground) to a plane surface. The bed knife shall be securely attached to the backing bar with either machine screws or rivets.

3.5 Drive wheels. The lawn mower shall have two drive wheels arranged so that the cutting reel will operate when either wheel is rotated in the forward direction. The wheel casting shall be cast aluminum with a fine finish on the teeth. The casting shall be carefully cleaned of all foreign material, metal chips removed and a coating of grease applied to the teeth prior to final assembly. The mowers shall have the wheel base fitted with either an oil impregnated sintered bronze bearing, or steel dimpled sleeve bearing packed with grease. The wheels shall be fitted with semi-pneumatic or solid, cushioned rubber tires. Tires shall be attached to the wheel in such a manner to prevent slippage. Plain wheels or plain wheels with slip over rubber tires will not be acceptable. The drive wheels shall be adjustable in height to a minimum of three positions on the side plates.

3.6 Pinion gear. The lawn mower reel shall be driven from both wheels, through the pinion gear with suitable ratchet action to release the reel when the wheels are stopped or moved in the rearward direction. Pinion gears shall be cast or sintered iron or steel with a fine finish on the teeth, and be impregnated with oil or greased prior to assembly. The ring gear and the pinion shall mesh fully and smoothly.

3.7 Pawls or pins. Pawls or pins shall engage the ratchet arrangement. They shall be steel or homogeneous alloy steel. If homogeneous, they shall have an average hardness of Rockwell C42 to C53, (see 4.4.4). If casehardened, depth shall be not less than 0.010 inch and they shall have a hardness number of 81 to 91 on the Rockwell 15-N scale.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

3.8 Roller and roller brackets. The machine shall have a seasoned hardwood, metal, plastic or rubber-covered one-piece or sectional roller, supported by adjustable roller brackets. The roller shall be not less than 1-1/2 inches in diameter and the overall length shall be not less than the width of cut minus 2 inches. The one-piece roller shall have a steel through shaft on which the roller is free to turn or steel end pins which shall be screwed or securely attached into the roller, with roller and pins rotating as one unit. Sectional rollers shall be free to rotate on a steel through shaft. The through shaft shall be not less than either 1/2 inch diameter steel tubing or 5/16 inch diameter steel bar stock minimum. The shaft shall be straight with no apparent curvature when rolled on a flat surface. The roller brackets shall be adjustable to at least three positions for raising and lowering the roller.

3.9 Frame. The side frames shall be rigidly secured in relation to each other by means of a tie rod or other structural member so designed as to permit easy removal of the reel assembly. The type II mower shall have a 1 inch diameter minimum straight shrub bar.

3.10 Handle and handle grip. The handle shall be constructed of steel or hardwood. The handle braces shall be constructed of steel and shall be attached to the handle by not less than two bolts with nuts. When wood is used, the handle grip shall be securely bolted to the handle through plates or brackets. When metal is used, the handle grip shall be securely fastened to the handle and shall be covered with rubber or suitable composition, except when the handle and handle grip is of a continuous one-piece construction forming a modified "U" shape. A "U" shaped handle shall be either chrome plated or other suitable means to prevent corrosion. The handle shall be of sufficient length, and angularly adjustable without the use of tools to allow an operator (tall or short) to push the mower in a comfortable walking position without bending. The handle shall meet the handle assembly test of 4.4.6.

3.11 Grass-catcher lugs. Lugs or other suitable provision shall be provided for mounting a grass catcher, when grass catcher is specified (see 3.14). The lugs may be integral with the roller brackets (see 3.8).

3.12 Lubrication. The lawn mower shall be fully lubricated prior to delivery. All bearings shall have suitable means for lubrication unless permanently packed with grease.

3.13 Shearing. Each blade on the reel shall shear one thickness strip of newspaper the entire length of the cutting blade when tested in accordance with 4.4.2. For the type I mowers there shall be no metal to metal contact between the reel blade and the cutting blade.

3.14 Grass catcher. When specified (see 6.2), mowers shall be furnished with grass catchers. They shall be either of sheet metal, woven wire, plastic, or metal-and-canvas construction. They shall be durable in service and shall be readily fastened to the lawn mower.

3.15 Cutting requirements. Rotating reel type mowers shall be capable of cutting heavy, thick-growth grass, 2 inches in height above the bed knife, in a single pass when operating under conditions herein specified. The cut shall be smooth and even for all heights of cut above the ground up to 1-3/8 inches. The type I, class A mowers shall make no fewer than 9.0 cuts per linear foot of travel and the type I, class B mowers shall make no fewer then 10.5 cuts per linear foot of travel. The type II, class A mowers shall make no fewer than 7.5 cuts per linear foot of travel and the type II, class B mowers shall make no fewer than 9.0 cuts per linear foot of travel.

3.15.1 Height of cut above ground. The height of cut shall be adjustable throughout a minimum range of 5/8 to 1-3/4 inches above the ground. The height of cut adjustment shall be firmly secured and shall not slip or get out of adjustment during the operation of the machine over rough terrain. The wheels shall be adjustable in height to allow the mower to remain essentially horizontal to the ground when changing cutting height (see 3.5).

3.16 Finish. All exterior metal surfaces shall be finished with a smooth durable weatherproof paint or lacquer to resist corrosion. Handle frames may be plated to resist corrosion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

3.17 Instructions and parts list information. The contractor shall furnish with each lawn mower either a booklet(s) or pamphlet(s) giving complete instructions for the operation, lubrication, adjustment, and care of the mower, and attachment units. A pamphlet listing all repair parts and parts source information shall also be furnished with each lawn mower.

3.18 Identification marking. Each lawn mower shall be marked in a plain and permanent manner with the model designator and name of the manufacturer or with a trademark of such known character that the source of manufacture may be readily determined.

3.19 Workmanship. The completely fabricated mower shall conform to the quality and grade product covered by this specification. The mower shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, rust, deterioration, or malformations. There shall be no sharp edges, burrs, or slivers. Workmanship shall be first class throughout.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. A random sample shall be selected from each lot in accordance with the provisions of MIL-STD-105 at inspection level II.

4.2.1 Sampling for dimension and tests. Sampling for dimension and tests shall be in accordance with MIL-STD-105, inspection level S-2, Acceptable Quality Level (AQL) of 4.0.

4.3 Examination.

4.3.1 Visual examination. The completely assembled end item shall be examined for defects in finish, construction, workmanship, and marking. Defects shall be as classified in table II. The inspection level shall be level II with an AQL of 4.0 percent for major defects and 6.5 percent for total defects expressed in terms of percent defective.

TABLE II. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not finished where required.	X	
	Type of finish not as specified, or blistered, peeled, chipped, or area of no film or thin film.	X	
	Any component that is painted or otherwise coated where serviceability may be affected.	X	
	Significant rust.	X	
	Negligible rust.		X
Type	Not type specified.	X	
Size	Not size specified.	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects (con.)

Examine	Defects	Classification Major	Minor
Construction and workmanship, general	Part missing or not as specified.	X	
Castings	Cracked, blowholes, pits, foreign material rough edges or other defects.	X	
Marking identification, instruction manual	Missing, incomplete, not legible, not as specified.	X	

4.4 Tests.

4.4.1 Gear housing openings. Clearances or openings in gear housings shall be determined by using a 3/32-inch-diameter metal rod. If the opening permits the passage of the rod end, the lawn mower does not comply with the requirements of 3.4.

4.4.2 Shearing. For type I mowers, each cutting reel blade shall be tested as follows: Turn the mower over and insert either a single thickness strip of newspaper approximately 2 inches wide or a .003 inch feeler gauge between the cutter bar and each reel blade to insure clear ance along its entire length. Turn reel slowly backward with wheels elevated. Adjust cutter bar so that the reel blades pinch the paper without cutting it. Paper should be able to be removed without tearing. After each reel blade has been tested, insert a strip of newspaper between the cutter blade and the reel blade perpendicular to the cutter blade. Rotate the reel by hand in the forward or normal cutting direction against the paper strip. The paper shall shear cleanly. The mower shall shear the paper the entire length of the cutting blade, for compliance with 3.13. For type II mowers each cutting reel blade shall be tested as follows: Adjust the cutting bar so each blade just touches the cutting bar across its entire length after the cutting bar has been adjusted and the reel rotated to check for a smooth uniform drag. Proceed using a newspaper strip as described above to test for cutting across the entire length of each blade. The paper shall shear cleanly for compliance with 3.13.

4.4.3 Mowing tests. Mowing tests shall be made on average smooth lawns or terraces. For the purpose of the mowing tests, to cut cleanly is defined as an even cut without waviness and which appears straight to the eye when observed from a distance of from 10 to 20 feet from the ends of each cut strip in accordance with 3.15.

4.4.4 Hardness. The hardness of the pawls, cutting reel blades, and bar knife shall be tested in accordance with ASTM Standard E 18 for compliance with 3.4.1 and 3.7.

4.4.5 Drive gears and pinion gears. Remove drive wheels and visually check drive wheel gear, pinion gear and pawls for compliance with 3.5 and 3.6. Measure reel shaft for compliance with 3.4.1.1.

4.4.6 Handle assembly test. The test shall consist of twisting the hand gripping bar clockwise, setting the mower on the right wheel with the left wheel clearing the floor. The handle shall be twisted counterclockwise until the right wheel clears the floor. This test shall be repeated 50 times and an inspection made. Any evidence of cracks or permanent deformation in the handle assembly shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.

4.4.7 Frequency of cut. The frequency of cut shall be calculated from the number of revolutions made by the reel to the number of complete revolutions made by the drive wheels. The ratio may be determined for the number of revolutions of the reel for each revolution of the drive wheel. The frequency of cut shall be calculated as follows:

$$F = \frac{R \times N}{C}$$

Where: F = Frequency (cuts per foot of travel).
R = Ratio of reel to wheel revolutions.
N = Number of reel knives.
C = Circumference of drive wheels (feet).

Mowers with a frequency of cut of less than specified shall be considered to have failed the frequency of cut requirements (see 3.15).

4.5 Rejected lots. Rejected lots may be resubmitted for examination and tests, provided that contractor has removed or repaired all nonconforming lawn mowers. Samples shall again be selected and inspection from such resubmitted lots to verify compliance with this specification. Rejected lawn mowers shall not be resubmitted for inspection without furnishing full particulars concerning previous rejections and measures taken to overcome the defects.

4.6 Examination of preparation for delivery. An examination shall be made to determine that the preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-1 and the AQL shall be 2.5 percent defective.

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Weight	Gross weight of shipping container is greater than permitted.

5. PREPARATION FOR DELIVERY

5.1 Preservation packing. Preservation packing shall be level A or Commercial (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation. All exposed metal surfaces shall be cleaned, thoroughly dried, and coated with a thin film of cold-application, solvent cutback corrosion preventive compound. The lawn mower shall be wrapped with aluminum foil or heavy duty flexible, greaseproofed, waterproofed, noncorrosive barrier material. The foil or barrier shall be loosely applied to the lawn mower and secured in place by taping, tying, or other suitable means. Projections, sharp edges, or other features of the lawn mower which may damage the barrier shall be cushioned.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:00Z
Check the source to verify that this is the current version before use.

5.1.1.2 Packing. Preserved lawn mowers shall be packed in nailed wood boxes or wood cleated plywood boxes. The boxes' construction shall be suitable for overseas shipment and open storage. The boxes shall be closed and strapped with flat steel strapping measuring not less than 1/2 x 0.020 inch. When the gross weight exceeds 200 pounds, each box shall be constructed with skids or a skid base to provide for access by forklift.

5.1.2 Commercial. Each lawn mower shall be preserved in accordance with the supplier's commercial practice and packed in a fiberboard box to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and to be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.2 Marking. Marking shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The mowers covered by this specification are intended for heavy-duty commercial use in the care and maintenance of lawns and grounds for which the Government is responsible.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type (see 1.2.1).
(c) Class (see 1.2.1).
(d) Size (see 1.2.2).
(e) When grass catcher is required (see 3.14).
(f) Levels of preservation and packing required (see 5.1).
(g) Civil or military marking as required (see 5.1).

6.3 Service guarantee. It is suggested that contracting officer include the following clause in the invitation for bid, contract or order:

"Service guarantee. The contractor guarantees the equipment against defective material, workmanship, and performance for a period of 1 year, said guarantee to run from date of acceptance of the equipment by the Government. The contractor agrees to furnish, without cost to the Government, replacement of all parts and material which are found to be defective during the guarantee period. Replacement material and parts shall be furnished to the Government at point of installation if installation is within the continental United States, or f.o.b. the continental United States port to be designated by the contracting officer if installation is outside of the continental United States. Cost of installation of replacement material and parts shall be borne by the Government".

U.S. GOVERNMENT PRINTING OFFICE : 1978 - 261-423/1213

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:00Z
Check the source to verify that this is the current version before use.